

Intelligent Student Grievance Redressal System

TEAM MEMBERS:

- Zoeb Gittham
- Megh Patel
- S Aakash
- Shreyas Srivastava

Project Guide : Mrs. Akruti Pandwal



Problem Statement

- Manual & inefficient

Paper forms and ad-hoc emails cause lost complaints and delays.

- Lack of transparency

No reliable tracking or status updates for students or staff.

- No prioritization

All complaints treated the same, critical issues overlooked.

- Delayed resolution

Slow routing and manual follow-ups extend resolution times.



Project Objectives



Centralized Management

Single source for submitting, tracking, and auditing complaints.



Secure Authentication

JWT-based login, role-based access control, and protected APIs.



Intelligent Classification

Hybrid AI identifies department, priority, and confidence scores.



Automated Routing & Tracking

Auto-route to relevant stakeholders and maintain audit trails.



System Overview



This flow ensures authenticated submissions, automated intelligence for triage, and a tracked lifecycle from creation to closure.

Module Division — Team Responsibilities



**Shreyas
Srivastava**
— Backend

Django backend,
REST APIs, DB
schema and core
business logic.



Megh Patel
— Security

JWT authentication,
role mapping,
protected endpoints
and admin config.



S Aakash
— Frontend

Responsive UI for
students and staff,
complaint
submission &
tracking views.

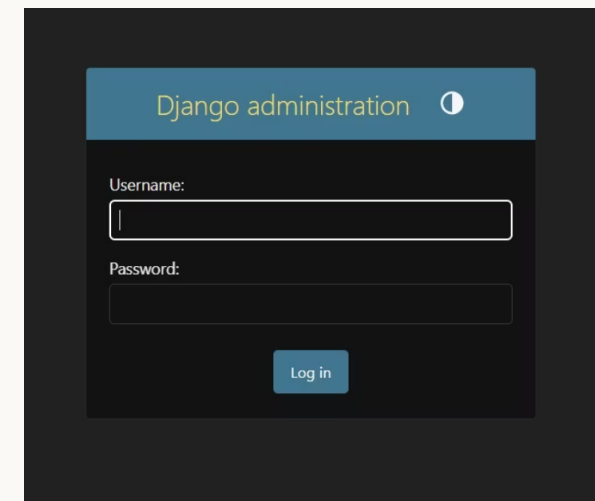
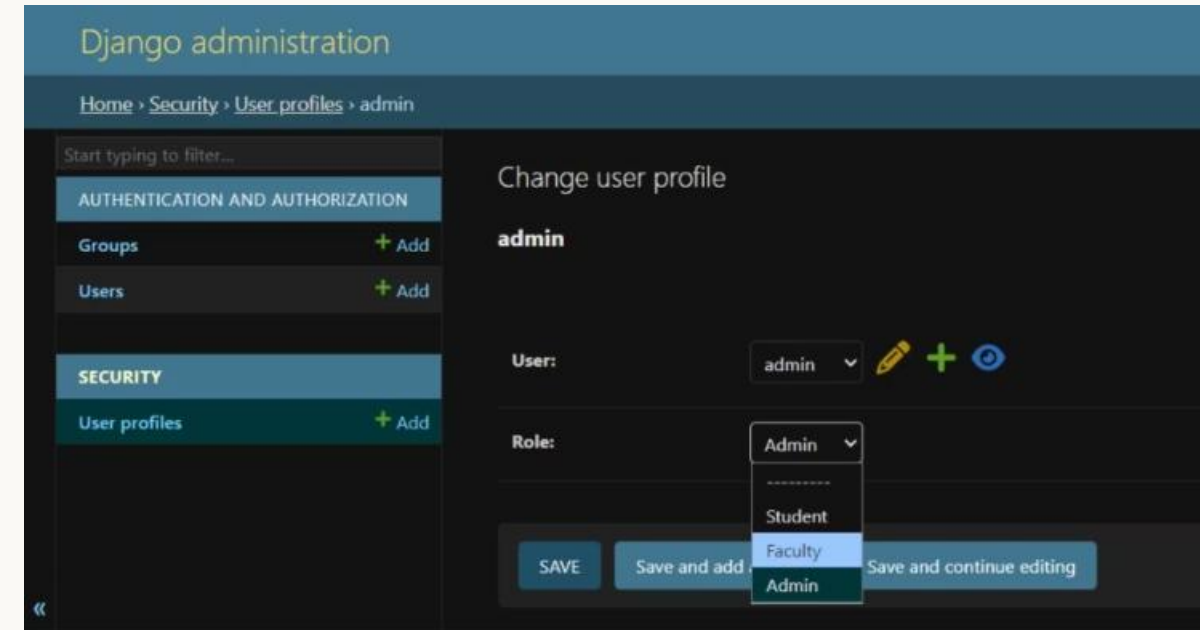


Zoeb Gittham
— AI Engine

Hybrid classification
engine: rule-based +
ML models, model
storage, and APIs.

Security Module — Implementations

- JWT Authentication — Implemented JWT-based auth for stateless, secure sessions.
- Secure Login API — Developed endpoints for login, token issuance, and refresh flows.
- Token Authorization — Header-based token validation and middleware for protected routes.
- Role-Based Access — UserProfile model maps roles: Student, Faculty, HOD, Admin — enforced in views.
- Admin Management — Configured Django Admin for user and complaint lifecycle oversight.



Security Module — Future Enhancements (You)

Multi-role Authorization

Granular permissions per role (Student, Faculty, HOD, Admin).

Anonymous & Encrypted Complaints

Optional encryption for sensitive reports, maintaining anonymity.

Hash-chain Tamper-Proofing

Append-only hash chain to ensure immutable complaint history.

2FA & Rate Limiting

Two-factor authentication and API rate limits to reduce abuse.

Audit Logs

Comprehensive activity logs for compliance and investigations.

AI Engine — Work Completed

- **Rule-Based Engine** — Keyword mapping, TextBlob sentiment analysis, dangerous keyword detection, and deterministic overrides.
- **ML Models** — TF-IDF vectorization + Multinomial Naive Bayes for department & priority; models saved as .pkl files.
- **Modular Design** — Pluggable AI modes (rule-based or ML), service-based architecture, config-driven switching.

The AI engine enables automatic, intelligent classification of complaints — returning department, priority, and a confidence score.

Analyze Complaint

POST /api/ai/analyze/

HTTP 200 OK
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept

```
{  
  "department": "hostel",  
  "priority": "Medium",  
  "confidence": 0.92,  
  "source": "ml"  
}
```

Create Complaint

POST /api/complaints/create/

HTTP 201 Created
Allow: POST, OPTIONS
Content-Type: application/json
Vary: Accept

```
{  
  "id": 3,  
  "title": "Attendance Issue",  
  "description": "My attendance is not updated for 2 subjects",  
  "department": "academics",  
  "priority": "Medium",  
  "ai_confidence": 0.92,  
  "ai_source": "ml",  
  "status": "Pending",  
  "created_at": "2026-02-25T05:52:07.996987Z"  
}
```


AI Engine — Future Enhancements



Continuous Learning

Models auto-retrain with new complaint data, ensuring accuracy and adaptability to emerging issues.



Contextual NLP

Deeper understanding of complaint nuances, sentiment, and user intent beyond keywords.



Predictive Insights

Identify recurring issues and potential bottlenecks proactively, enabling preventative measures.

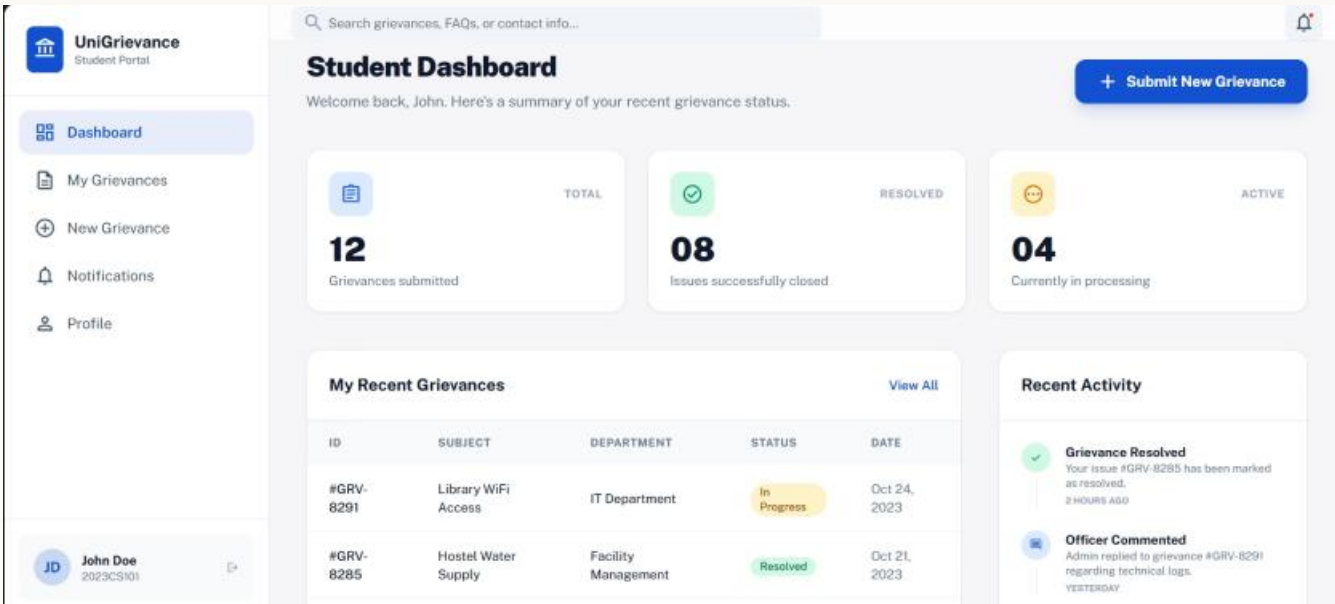
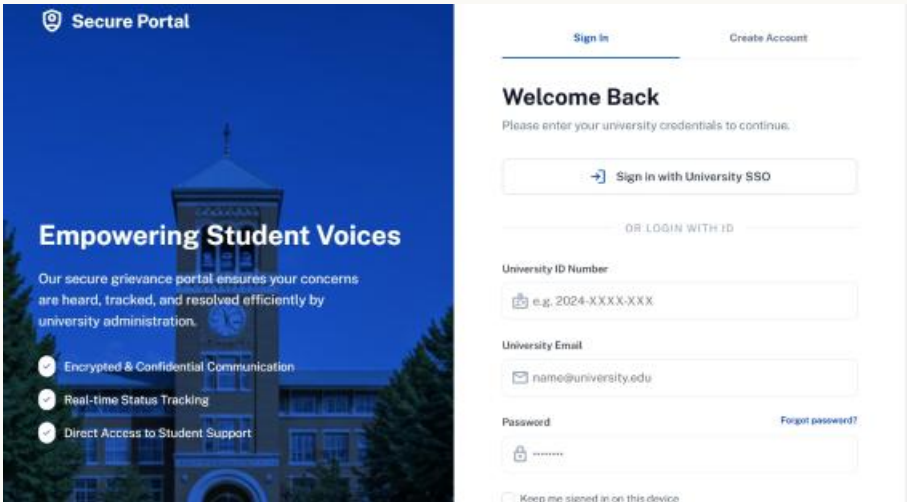


Resolution Suggestions

Auto-suggest solutions to students and staff by integrating with knowledge bases and FAQs.

Frontend Module — Work Completed

- **Login Interface Designed**
 - Separate UI for Student/Admin login
 - Clean and structured form layout
 - Input validation at UI level
- **Student Dashboard UI Created**
 - Dashboard layout structured
 - Navigation panel designed
 - Space allocated for grievance status display
- **Grievance Submission Page Designed**
 - Form for entering grievance details
 - Fields structured properly (title, description, category if required)
 - User-friendly interface for complaint submission
- **UI-Backend Integration (Partial)**
 - Grievance form connected to backend API
 - Data successfully sent to Django server



Frontend Module — Future Enhancements



Full Authentication Handling

- Proper session management for user states
- Role-based redirection for Student/Admin



Admin Dashboard UI

- Comprehensive view of all submitted grievances
- Advanced filter options (Department / Priority / Status)
- Intuitive status update controls for administrators



Grievance Tracking Interface

- Students can view submitted complaints
- Real-time current status updates
- Visibility of AI-predicted category & priority



Advanced Enhancements

- Dark/light mode toggle for user preference
- Continuous improvement of user experience design



UI Improvements

- Responsive design optimization for all devices
- Seamless loading indicators & alerts
- Clear success/error notifications

Current Progress of Backend Module

1. Project Structure Setup (Django)

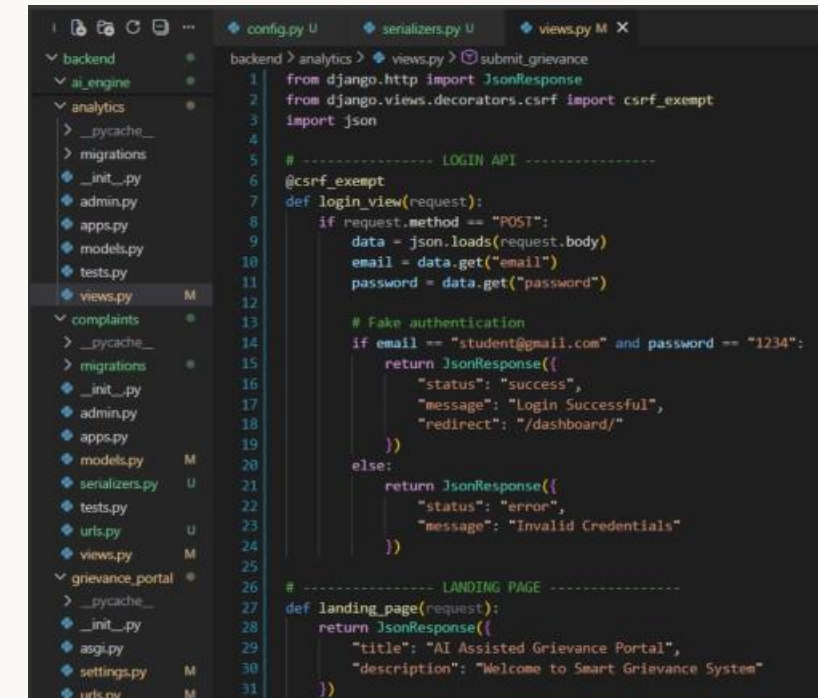
- Django project initialized
- Apps configured properly
- Basic settings configured

2. Routing Configuration Completed

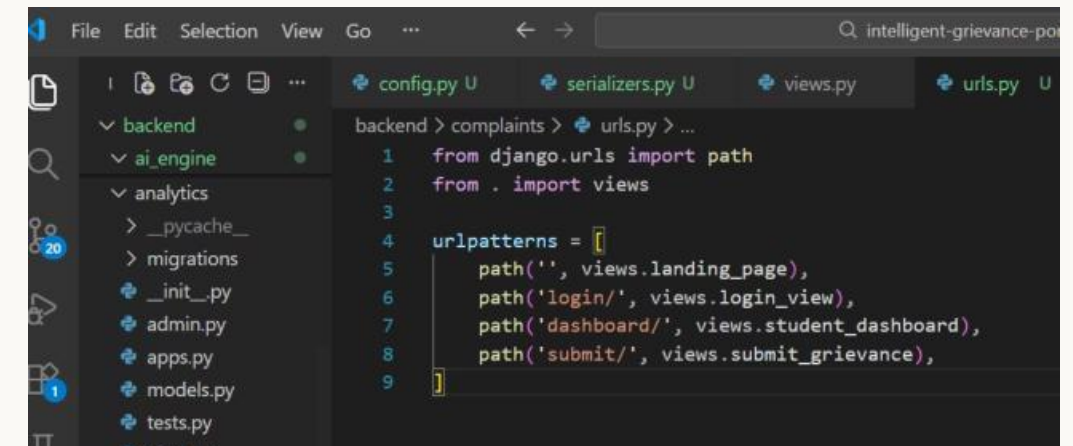
- URLs mapped for: Login page, Student dashboard, Grievance submission page
- Proper navigation between frontend pages through backend routing

3. Views Implemented

- Backend view functions created for: Rendering login page, Rendering dashboard, Handling grievance submission requests
- Basic request-response cycle working



```
1 from django.http import JsonResponse
2 from django.views.decorators.csrf import csrf_exempt
3 import json
4
5 # ----- LOGIN API -----
6 @csrf_exempt
7 def login_view(request):
8     if request.method == "POST":
9         data = json.loads(request.body)
10        email = data.get("email")
11        password = data.get("password")
12
13        # Fake authentication
14        if email == "student@gmail.com" and password == "1234":
15            return JsonResponse({
16                "status": "success",
17                "message": "Login Successful",
18                "redirect": "/dashboard/"
19            })
20        else:
21            return JsonResponse({
22                "status": "error",
23                "message": "Invalid Credentials"
24            })
25
26 # ----- LANDING PAGE -----
27 def landing_page(request):
28     return JsonResponse({
29         "title": "AI Assisted Grievance Portal",
30         "description": "Welcome to Smart Grievance System"
31     })
```



```
1 from django.urls import path
2 from . import views
3
4 urlpatterns = [
5     path('', views.landing_page),
6     path('login/', views.login_view),
7     path('dashboard/', views.student_dashboard),
8     path('submit/', views.submit_grievance),
9 ]
```

Backend Module: Next Steps & AI Integration

Our backend development is progressing rapidly. Here are the key areas we're focusing on for upcoming completion and integration.

1

Database Integration
Connecting models to store grievance records including student details, complaint text, AI predictions (department, priority), and status permanently.

2

Authentication System
Implementing user registration, login validation, session management, and robust role-based access for students and administrators.

3

Admin Functionalities
Developing features to retrieve, update (Pending, In Progress, Resolved), filter, and sort all grievances for efficient management.

4

AI Engine Integration
Integrating the AI module to process grievance text, receive predicted department and priority, and successfully handle these responses.

5

Deployment Preparation
Configuring environment settings, production setup, and critical security improvements for system readiness.

Technologies Utilised in Our Project

Our system leverages a robust stack of modern technologies to ensure efficiency, scalability, and intelligence.



Frontend

- HTML5, CSS3, JavaScript for UI
- Bootstrap for responsive design



Database

- SQLite for development
- MySQL for production scalability



Integration

- REST API with JSON communication
- Seamless frontend-backend data exchange



Backend

- Python as core language
- Django Framework for robust APIs



AI Module

- Python-based text processing
- Keyword classification & priority detection



Dev Tools

- VS Code for coding
- Git for version control

Conclusion

Our Intelligent Student Grievance Redressal System is poised to transform how student concerns are managed, creating a more efficient and transparent environment.

Streamlined Operations

Automated processes and intelligent routing significantly reduce manual effort and resolution times.

Enhanced Transparency

Students gain clear visibility into the status and progress of their complaints, fostering trust and satisfaction.

Data-Driven Insights

AI-powered classification provides valuable data for identifying recurring issues and improving institutional policies.

